

Keeping Pace: A Practitioner-Focused Review of Pacing Strategies in Running

J. Luke Pryor, PhD, ATC, CSCS,¹ Evan C. Johnson, PhD,² Hillary A. Yoder, MS,³ and David P. Looney, PhD, CSCS⁴
¹Department of Exercise and Nutrition Sciences, Center for Research and Education in Special Environments, University at Buffalo, Buffalo, New York; ²Division of Kinesiology & Health, University of Wyoming, Laramie, Wyoming; ³Department of Kinesiology, University of Alabama, Tuscaloosa, Alabama; and ⁴United States Army Research Institute of Environmental Medicine (USARIEM), Natick, Massachusetts

ABSTRACT

The purposeful distribution of speed, power, or energy is termed as the pacing or pacing strategy and is recognized as a key determinant in optimal run performance. There is no agreement on the best pacing strategy for all runners and race types. Thus, the challenge posed to runners and practitioners is pacing strategy selection and in-race adherence. This review briefly discusses pacing strategy types and selection considerations. More importantly, we overview factors influencing pacing and translate key findings from research into useable evidence-based recommendations for pacing strategy preparation and adherence during competition.

INTRODUCTION

Pacing is the purposeful distribution of speed, power, or energy during exercise. The thought process behind pacing (i.e., strategy) is recognized as an important element in endurance performance optimization. The challenge posed to runners is to avoid premature energy depletion resulting in fatigue or finishing a race

with excess energy resources. How runners regulate exercise intensity and the decision-making process involved in investing limited energy reserves during a race is debated. Several theories attempt to explain the regulatory mechanisms underpinning the process of pacing and pacing decision-making (Table). Generally, pacing is a dynamic and complex incorporation of physiological and psychological cues integrated consciously and subconsciously in an anticipatory manner to achieve optimal performance. Further discussion of mechanisms underpinning pacing is outside the scope of this practitioner-focused review. Rather, this review aims to translate key findings from research into useable evidence-based pacing strategy recommendations for runners and practitioners. To this end, we overview various types of pacing strategies and considerations for selection followed by discussion of influential factors with recommendations for preparation and adherence.

PACING STRATEGY OVERVIEW AND SELECTION

There is not unanimous agreement on the “best” pacing strategy for all runners and race types, in part, because pacing is a complex process influenced by a number of factors (Figure 1). Figure 2

illustrates the 5 general pacing strategies: even, positive (fast-slow), negative (slow-fast), variable, and terrain-dependent. When considering which strategy to use, some argue event duration or distance should dictate selection, particularly in time-trial type competitions. An “all-out” strategy in events <60 seconds (e.g., 100 m, 200 m, or 400 m) is indicated by mathematical pacing models mainly addressing physiological and biomechanical components (14). A positive (fast-slow) pacing strategy is best for 1- to 2-minute events (i.e., 800 m) (27,84). However, an even pacing strategy seems to be optimal in events of 2 minutes to 24 hours based on mathematical models (76,87), men’s track world-records (84), and ultraendurance events (29,38).

However, most real-life competitions are not characterized by time-trial type competitions (28). Runners must balance human-environment interactions (e.g., weather, competitors, nutrition/hydration, etc.) and in-race tactical decisions into the regulation of exercise

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energy expenditure; even; negative; positive; endurance; time-trial; regulation

Address correspondence to Dr. J. Luke Pryor, lprior@buffalo.edu.

Table Various theories or models that attempt to explain the regulatory mechanisms underpinning the decision-making process of pacing	
Theory or model	Author
Teleoanticipatory	Ulmer (85)
Central governor	Noakes et al. (57)
Perception-based	Tucker (83)
Pacing awareness	Edwards and Polman (18)
Psychobiological	Marcora and Staiano (44)
Dual process	Micklewright et al. (50)
Heuristic decision-making	Renfree et al. (66)
Ecological-psychological	Smits et al. (75)

intensity (28,75), suggesting that optimal pacing requires additional considerations beyond distance/duration. Evidence from laboratory studies indicate distance runners adopt an even-pacing strategy that minimizes physiological, perceptual, and metabolic disturbances, ultimately improving exercise likability and maximizing performance (1,4,78,79). On the other hand, endurance athletes tend to adopt

a positive or variable-pacing strategy in competition (2,39,89). These findings led some practitioners to suggest a negative pacing strategy to avoid late-race fatigue (23). Finally, a terrain-dependent pacing strategy may be best for race courses with undulating topography (31,61,81).

The inconsistent recommendations among practitioners, pacing models, and laboratory and observational

studies underscore the fact that there is not one “optimal” pacing strategy for all runners and events. Rather, this should be taken as a beginning for exploring pacing strategies in various individuals and events. Leveraging pacing in competition requires proper strategy selection with an understanding of factors that could influence in-race pacing decision-making and overall adherence (36,50,75).

FACTORS INFLUENCING PACING AND PRACTICAL RECOMMENDATIONS

PRACTICE

A pacing strategy should be identified and integrated early into training programming. Pacing is a self-regulation skill that, like most athletic skills, must be practiced and crafted (19). Previous race experience and/or competition emulation is essential for developing skills that optimize pacing (6,51,83). For example, experienced endurance athletes show effective information seeking behavior (6) and attentional focus on task-related sensory information (e.g., physical and mental exertion and pain) (55), which guide pacing

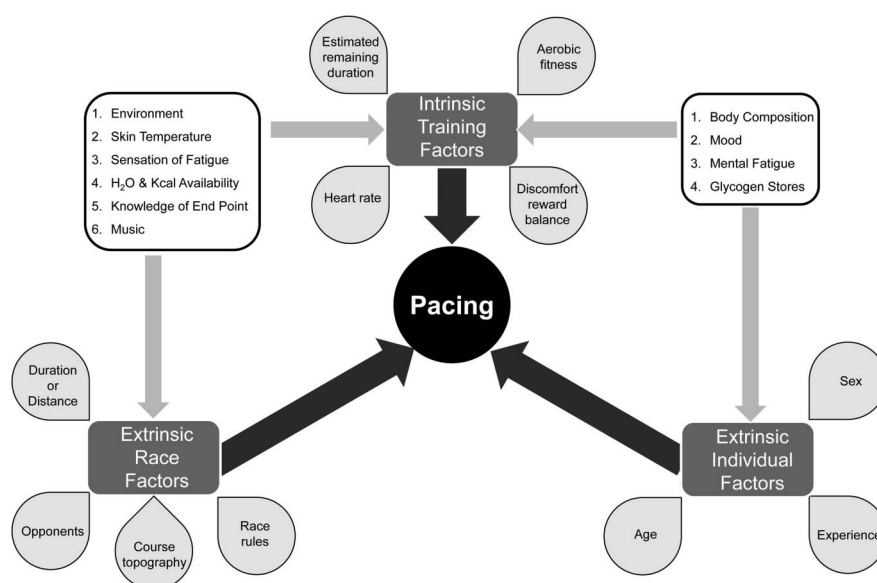


Figure 1. Factors believed to influence pacing. Factors influencing pacing strategy and/or decision-making consistent throughout all theories and models (Table) include the following: sensations of fatigue, perceived level of effort, knowledge about the end point, estimated remaining distance or duration, and the willingness to succumb to discomfort in anticipation of reward (36,66).

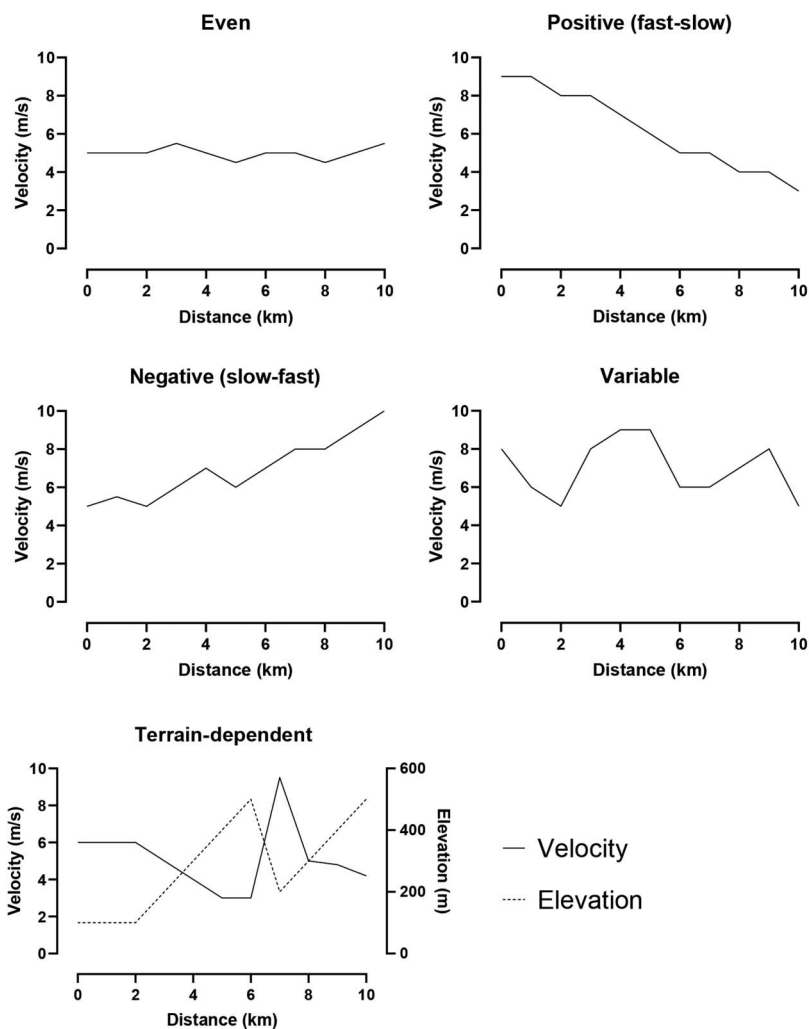


Figure 2. General types of pacing strategies. Energy distribution is commonly described in practical terms such as run velocity (mph or kph) or pace (min/mile) as is it difficult to quantify the total energy reserves of a runner and energy expenditure (e.g., oxygen consumption) outside of a laboratory setting.

decision-making. Ideally, coaches and runners should practice foreseeable in-race situations and determine appropriate decision-making options based on available information congruent with the predetermined pacing strategy. Such actionable information might include distance covered or remaining, major terrain changes, feelings of effort or fatigue, and in-race tactics of opponents.

NUTRITION

Nutrition (encompassing hydration) plays a vital role in endurance

performance and pacing with growing influence as race duration/distance increases (12,61,65,68). Both dehydration (5) and low glycogen stores (12,65) result in an earlier and more rapid decline in run velocity or power output, with the potential absence of an end spurt (41). In addition, greater cardiovascular strain occurs as dehydration progresses (cardiac drift) (54). Cardiovascular strain seems to influence effort perception and peak oxygen uptake, which is important as exercise intensity selection is partly dependent on perceived effort (22,70,83).

The ability to sustain in-race glycogen stores and hydration status can vary drastically based on intrinsic (i.e., athlete choice) and extrinsic factors (e.g., event provision and distance between aid stations) factors. Minimizing dehydration and glycogen depletion is particularly vital in hot/humid weather and/or long-duration events when these conditions are most likely to occur (21,71). Practicing prerace and in-race nutritional strategies is recommended (68).

High-carbohydrate (CHO) diets, CHO loading, and high in-race CHO intake rates (30–90 g CHO/hr) are evidence-based recommendations inversely related to finish times in longer events (59,68) despite potential gastrointestinal complaints (59). CHO intake can occur in a variety of manners, without one clearly defined best option. For example, runners can pre-mix a CHO beverage containing 90 g of CHO/L and attempt to consume 1 L over the course of an hour. In events with aid stations, the runner may instead choose to take in large quantities of CHO in a single feeding. Longer finish times, larger elevation changes, and runners with higher oxygen consumption rates or body mass require greater CHO consumption rates. The metering of CHO is partially dependent on the athlete's goal finish time. This does not mean that faster runners always need more CHO per event or per hour. Outside of ketogenic athletes, in-race fat and protein consumption does not enhance performance (88), and it is unknown whether a ketogenic diet influences pacing.

Fluid intake composition and rate should be determined and practiced before competition (48,63). For recreational runners and competitions in mild weather, drinking to thirst is likely adequate (24). However, for heavy sweaters, competitions in the heat, and those striving to maximize performance, minimizing dehydration to <2% body mass loss is warranted and should be accomplished by matching fluid intake to sweat rate (48). Caffeine consumption (3–6 mg/kg) reduces

effort perception and enhances triglyceride oxidation sparing glycogen which, in part, enhances aerobic performance (17,69) and potentially pacing fidelity.

Simple calculations and predictive equations are useful for determining general nutritional and fluid requirements to delay premature fatigue. For example, level running energy expenditure can be roughly estimated as $1 \text{ kcal} \cdot \text{kg body mass}^{-1} \cdot \text{km}^{-1}$ regardless of speed (46). A more elaborate equation is needed to determine energy costs of running on uphill and downhill slopes (53). Body water losses during running can be estimated from energy expenditure and pre-post change in body mass with a simple equation: $\Delta \text{Body Water} = \Delta \text{Body Mass} - 0.20/\text{kcal}$ (10). More accurate predictions of sweat loss require additional information (clothing, equipment, and environmental conditions) (60).

ENVIRONMENTAL CONDITIONS

Moderate-to-extreme environmental conditions increase the physiological and psychological strain incurred during running, challenging pacing strategy fidelity. The consequence of added environment stress on pacing may be manifest by earlier and greater decline in run velocity with or without marked variation. This section overviews the impact of heat, humidity, altitude, and wind on pacing.

Heat and humidity. Hot and/or humid conditions evoke earlier and larger declines in exercise intensity, as well as greater variation in pacing versus temperate conditions (13,20,58,64,82). Early pace modifications in the heat are likely due to increased thermal sensations related to elevated skin temperatures and then later cardiovascular strain, subsequently inflating perceived effort above expected levels (22,72). Thus, managing rapid increases in early race skin temperature and thermal sensations while minimizing cardiac drift in latter stages of hot weather races can aid in pacing strategy adherence and performance. Pre-event cooling strategies (e.g., cooling garments, ice slurry ingestion, and cold water immersion) seem

ergogenic during hot weather competitions by attenuating early sensations of hotness and skin or core temperature (11,63). Self-dousing extra water on the face may lower thermal discomfort and perceived effort (56). Heat acclimation improves endurance performance in the heat with longer protocols (10–14 days) showing the greatest benefit to both performance and maintenance of exercise intensity (64). Although practical recommendations for heat acclimation in team and individual sports are available (62), runners without time to fully heat acclimate may find benefit from even one day of heat familiarization (73).

Altitude. Altitude changes alter pacing and performance, especially in ultra-endurance races that seek to traverse varied terrain. As with heat-stress, hypoxia conditions result in earlier and greater declines in exercise intensity versus normoxia conditions (13,58). As exercise progresses in hypoxic conditions, the rate and capacity for aerobic energy contribution decrease with equivalent increases from anaerobic capacity to sustain pace. When this metabolic shift and associated higher sensory feedback (higher blood lactate and discomfort) integrated within the brain, the motor cortex reduces central motor drive, lowering the energy expenditure rate as a result of low oxygen availability (13) to better match expected versus perceived effort.

Altitude training mitigates the added hypoxia stress of high-altitude races, improving endurance performance and pacing profiles. Athletes residing at higher elevations may realize a physiological benefit when racing at a lower altitude (40), although they may not need to adjust their pacing strategy. To prevent late race performance reductions in nonacclimatized athletes in races with anticipated oppressive environmental conditions, coaches and runners may consider a negative (slow-fast) pacing strategy or at least starting at a slower pace and adjusting goal finish time (82).

Wind. Wind can significantly influence cycling velocity and pacing, but

because run velocity is slower, the effects of wind on run performance and pacing are comparatively attenuated. However, when attempting to maximize all potential sources of performance enhancement, drafting may be worthwhile as modeled by those predicting sub-2-hour marathon times (3,76) or if headwinds exist.

COURSE TOPOGRAPHY

Course topography has deterministic effects on run velocity, power output, and pacing strategy (31,61,81). The physiological demands of sustaining any given pace uphill is greater than flat ground running. Therefore, runners must reduce pace to maintain a physiological steady state; alternatively, sustaining pace is at the cost of elevated physiological and psychological disturbances. Conversely, downhill running enables increased velocity without linear increases in physiological demand due to gravity.

Oscillations in velocity in opposition to elevation (slower on uphill and faster on downhill) to sustain a physiological steady state are consistent with terrain-dependent pacing (31,61). The acceptable degree of run velocity variation during even, variable, and terrain-dependent pacing is debated. We (61) and others (2,3,67) purport an oscillation within 5% is satisfactory, but values up to 20% seem acceptable in cycling events with major elevation changes (61). Within this pacing strategy, velocity variation is dependent on several factors including course topography, duration, environmental conditions, in-race tactics, and exercise mode among others.

Practicing pacing on the actual race course is preferred, but not always possible. In such cases, driving or viewing the race course beforehand may aid in a better understanding of course topography and define information-seeking objectives (mile markers, hills, structures, etc.). Emulating race elevation changes on a treadmill seems ineffective as treadmill pacing differs from above ground running (26) likely because of the conscious requirement to adjust

speed or elevation. At minimum, emulating major elevation changes and distances of a race on alternate above-ground routes is practical. Terrain-dependent pacing seems worthwhile in longer events (31,61), but its utility in shorter events is unknown.

THE COMPETITION

When exploring the pacing strategy and behavior, it is beneficial to incorporate and practice the influence of opponents and opponent-based tactical positioning and expectations (36). An audience or competitor(s) usually results in better performance, a phenomena known as social facilitation (94). Racing against opponents enhances physiological and psychological aspects yielding faster run velocities without influencing effort perception (37,92).

However, opponents can also influence pacing decision-making (49,80). In several middle- and long-distance events, the best endurance athletes adopted positive pacing whereby the fast initial pace cannot be sustained by most subelite athletes (25,39,67,77). Subelite runners must decide to adopt the group pace or not. The perception of approaching or getting behind an opponent is usually (49) but not always (61) actionable information and highly motivational (49,61). Often, runners consciously and/or subconsciously deviate from their pacing strategy to remain within the group (80), consequentially adopting an unsustainable pace. Such a decision may not be perceived as hazardous due to large untapped energy resources (52) and negligible changes in effort perception due to the presence of opponents (37,91). Thus, a major internal cue of energy distribution (effort perception) may go unheard until greater physiological and psychological perturbation is recognized. If this unsustainable pace continues, premature fatigue sets in, leading to pacing strategy derailment and poor performance. For example, marathoners starting at their intended race pace sustained an even pace and performed better than those selecting an initial pace faster than their personal best time (67).

Discussion of the pacing strategy to this point has assumed that a runner's goal is to finish in the fastest time possible. In head-to-head competitions, finishing in the fastest time possible (with optimal pacing) may be irrelevant as long as a runner crosses the finish line first. Thus, pacing behavior may be influenced by in-race tactics and adjustments evoked in response to competitors. For example, in half (25) and full marathons (67) and Olympic triathlons (89), spontaneous group synchronization of velocity and movements, occurred mediating pacing behavior. In both 800- and 5,000-m events in the Sydney Olympics, the runner with the highest average velocity actually lost due to adverse tactical positioning on outer lanes of the track (32). Understanding how a runner responds to competitors and their tendencies could assist coaches in training preparation, in-race decision-making, and pacing strategy selection.

Because approaching or getting behind an opponent is usually actionable information, emulating such scenarios in practice aids in solidifying tactical pacing decision-making. Addressing the "fast-start" phenomena in novice runners is highly recommended as experience mitigates this response (41). With terrain-dependent pacing, it may be initially challenging for some runners to accept competitors passing them on uphill sections. This is a necessary response to sustain physiological responses within accepted thresholds. Therefore, reminding runners that they may catch up or exceed these opponents on downhill sections is warranted. To leverage social facilitation throughout the race, it may be advantageous to identify rivals or partners with similar abilities and run with them (25), especially in novice runners. Care should be taken to ensure race pace is comparable with training intensities.

COGNITIVE ASPECTS

Pace regulation requires constant evaluation and integration of a runner's perceived versus expected status on a number of psychobiological variables (Figure 1) (44,75). Interpretation of

these variables leads to pace modifications aimed at better alignment of the perceived and expected relationships. This persistent conscious/subconscious evaluation over time can alter cognition (90) and may precipitate mental fatigue, especially during taxing or prolonged races. Mental fatigue is associated with poor endurance performance and pacing strategy fidelity (8,86). In addition to mental fatigue, information gained throughout a race and how a runner perceives, integrates, and ultimately acts on this information is an essential determinant of pacing behavior (36).

Developing mental skills (imagery, self-talk, attentional focus, and goal setting) may attenuate mental fatigue by modulating exercise likability, perceived effort, attentional focus, and/or task engagement rather than impacting physiological processes (7,45,47,55). Elite marathoners tend to associate sensations (pain, discomfort, and cadence) of running to guide pace modifications, whereas novice marathoners dissociate from these sensations (55). The recreational runner could leverage this learned skill by identifying their most effective inward (e.g., breathing rhythm and cadence) and outward (e.g., mile markers and competition) attentional foci and alternate between them to adjust pacing in perspective of their mental state, physical condition, and current situation. Music can be motivational and stimulating, positively influencing both likability and attentional focus (43). The ergogenicity of music is related to matching the beat to cadence or selecting enjoyable music drawing attentional focus away from other internal/external cues (33). Although music may be an acceptable aid for recreational running, it is banned in major competitions and may be counterproductive for runners as there is potential for dependence and divulgence from associative cognition (55).

WEARABLE TECHNOLOGIES

Pacing optimization may require the use of wearable technologies that provide instantaneous feedback. Global positioning system devices measure run velocity and distance and can be

linked with topographical databases to determine elevation changes and running surface gradients (35). Heart rate monitoring is an inexpensive, time-efficient, and noninvasive approach to regulating training intensity (9). In addition, minute-by-minute heart rate measurements can be used to calculate real-time estimations of core body temperature (42). Electronic foot contact monitors provide further insight on force generation per running stride (30). Consequently, runners have many options for real-time tracking of exercise intensity and distance at their disposal.

TRAINING-INDUCED FATIGUE

Training has been shown to result in adopting a more theoretically optimal “even”-pacing strategy (34), in line with pacing profiles of elite long-distance runners (67,84). However, 2 studies show that 1–4 weeks of intensified training impair planned pacing (74,93). In both studies, after 2–4 days of recovery, pacing profiles returned to preintensified training patterning. A proper taper before major endurance events can optimize training adaptations, recovery, and pacing strategy fidelity.

SEX AND AGE

Because male runners tend to reduce run velocity (negative pacing) throughout marathon (16) and 10-k races (15) more than women, coaches and male runners should make concerted efforts to address this trend. Older and likely more experienced runners exhibit less slowing over a race (15,16) consistent with the thought that pacing is a skill to be practiced and crafted over years of experience (19).

SUMMARY

There is not a single “optimal” pacing strategy for all runners and events. Rather, consideration of factors known to influence pacing is required to develop and use strategies to leverage pacing in competition. Addressing these factors in training and practicing a planned pacing strategy is recommended. In-race adaptation(s) to

a planned pacing strategy in response to new information is important for overall adherence and successful race outcomes.

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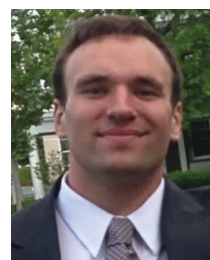
J. Luke Pryor is a clinical assistant professor of exercise science at the University at Buffalo.



Evan C. Johnson is an assistant professor of kinesiology at the University of Wyoming.



Hillary A. Yoder is a doctoral student in kinesiology at the University of Alabama.



David P. Looney is a research physiologist for the Biophysics and Biomedical Modeling Division at the United States Army Research

Institute of Environmental Medicine (USARIEM).

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